ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain asymptotic notations (Big-O, Omega, Theta). Write an algorithm for Circular Queue insertion and deletion. [14 Marks]

[OR]

Q.1 (Alt) Compare Singly, Doubly, and Circular Linked Lists. Write an in-place list reversal algorithm. [14 Marks]

--- UNIT II ---

Q.2 Describe infix to postfix expression conversion. Solve Tower of Hanoi recurrence relation. [14 Marks]

[OR]

Q.2 (Alt) Explain Double-Ended Queue (Deque) and Priority Queue array vs heap representations. [14 Marks]

--- UNIT III ---

Q.3 Define Binary Search Tree (BST). Explain AVL tree rotations (LL, RR, LR, RL) with examples. [14 Marks]

[OR]

Q.3 (Alt) Explain B-Trees of order 3 and Huffman Coding algorithm with prefix-free property. [14 Marks]

--- UNIT IV ---

Q.4 Compare BFS and DFS graph traversals. Apply Dijkstra's Single Source Shortest Path algorithm. [14 Marks]

[OR]

Q.4 (Alt) Explain Prim's and Kruskal's algorithms for finding Minimum Spanning Tree. [14 Marks]

--- UNIT V ---

Q.5 Explain Quick Sort partition strategy and average-case analysis. Discuss Hash collision techniques. [14 Marks]

[OR]

Q.5 (Alt) Explain Merge Sort with recursion tree and Heap Sort algorithm with max-heap construction. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****